

Brutal Practice Paper B11 - Light, Data Handling, Wastewater Story & The Triangle

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. Two plane mirrors are first set at an angle of 30 degrees, then opened out to 60 degrees. Using the rule that the number of images equals (360 divided by the angle) minus 1, how many fewer images form after they are opened out? [\[Light\]](#)

- (A) 2 fewer
- (B) 11 fewer
- (C) 6 fewer
- (D) 16 fewer
- (E) 4 fewer
- (F) 5 fewer
- (G) 7 fewer

2. The mean of five numbers is 18. If the number 30 is taken out of the set, what is the mean of the remaining four numbers? [\[Data Handling\]](#)

- (A) 12
- (B) 14
- (C) 20
- (D) 16
- (E) 22.5
- (F) 18
- (G) 15

3. A treatment plant receives 60,000 litres of sewage in which 5% is solid waste. How many litres of solid waste enter the plant? [\[Wastewater Story\]](#)

- (A) 3000 litres
- (B) 5000 litres
- (C) 300 litres
- (D) 30000 litres
- (E) 1200 litres
- (F) 12000 litres

4. The three angles of a triangle are in the ratio 2 : 3 : 4. What is the measure of its largest angle? [\[The Triangle & its Properties\]](#)

- (A) 90 degrees
- (B) 70 degrees
- (C) 80 degrees
- (D) 40 degrees
- (E) 120 degrees
- (F) 60 degrees
- (G) 100 degrees

5. A coin lies 8 cm in front of a plane mirror. The mirror is then slid 3 cm farther away from the coin. What is the new distance between the coin and its image? [\[Light\]](#)
- (A) 22 cm
 - (B) 5 cm
 - (C) 11 cm
 - (D) 14 cm
 - (E) 16 cm
 - (F) 19 cm
6. In a test, 30 students of Class A scored a mean of 60, while 20 students of Class B scored a mean of 70. What is the mean score of all 50 students together? [\[Data Handling\]](#)
- (A) 64
 - (B) 68
 - (C) 130
 - (D) 66
 - (E) 62
 - (F) 63
7. After sewage stands quietly in a sedimentation tank, the solids that settle at the bottom are removed. What is this settled material called and what is it often turned into? [\[Wastewater Story\]](#)
- (A) clarified water, which is drunk
 - (B) scum, which is sold as fuel oil
 - (C) activated water, which is bottled
 - (D) sand, which is used in building
 - (E) chlorine, which kills germs
 - (F) sludge, which is turned into biogas and manure
8. An exterior angle of a triangle is 110 degrees. One of the two opposite interior angles is 45 degrees. What is the other opposite interior angle? [\[The Triangle & its Properties\]](#)
- (A) 45 degrees
 - (B) 110 degrees
 - (C) 155 degrees
 - (D) 70 degrees
 - (E) 65 degrees
 - (F) 35 degrees
 - (G) 55 degrees
9. A wall clock seen in a mirror appears to show 3:20. How many minutes after the real time will it be 9 o'clock? [\[Light\]](#)
- (A) 20 minutes
 - (B) 60 minutes
 - (C) 220 minutes
 - (D) 160 minutes
 - (E) 40 minutes
 - (F) 80 minutes
 - (G) 100 minutes

10. Four of five numbers are 20, 28, 15 and 31. If the mean of all five numbers is 24, what is the fifth number? [\[Data Handling\]](#)

- (A) 26
- (B) 30
- (C) 22
- (D) 94
- (E) 24
- (F) 120
- (G) 18

11. In an aeration tank, a blower pushes in 250 litres of air every minute for 90 minutes. How many litres of air are blown in altogether? [\[Wastewater Story\]](#)

- (A) 225000 litres
- (B) 2250000 litres
- (C) 11250 litres
- (D) 22500 litres
- (E) 2250 litres
- (F) 340 litres
- (G) 18000 litres

12. A right-angled triangle has its two shorter sides equal to 9 cm and 12 cm. What is its perimeter? [\[The Triangle & its Properties\]](#)

- (A) 15 cm
- (B) 36 cm
- (C) 21 cm
- (D) 225 cm
- (E) 30 cm
- (F) 45 cm

13. The word CHOICE is held up to a plane mirror. Counting only the letters that look exactly the same after the left-right flip, how many of the six letters are unchanged? [\[Light\]](#)

- (A) 0
- (B) 3
- (C) 1
- (D) 6
- (E) 5
- (F) 2
- (G) 4

14. Find the median of the data 7, 3, 9, 12, 5, 8. [\[Data Handling\]](#)

- (A) 6
- (B) 7.5
- (C) 9
- (D) 7
- (E) 8.5
- (F) 8
- (G) 12

- 15.** A treatment process removes 80% of the contaminants in its first stage, then 75% of whatever remains in its second stage. What percentage of the original contaminants is still left? [\[Wastewater Story\]](#)
- (A) 15%
 - (B) 20%
 - (C) 10%
 - (D) 25%
 - (E) 55%
 - (F) 5%
- 16.** In an isosceles triangle the angle between the two equal sides, the vertex angle, is 40 degrees. What is the size of each equal base angle? [\[The Triangle & its Properties\]](#)
- (A) 100 degrees
 - (B) 80 degrees
 - (C) 140 degrees
 - (D) 40 degrees
 - (E) 70 degrees
 - (F) 50 degrees
 - (G) 20 degrees
- 17.** A ray of light strikes a plane mirror and the angle between the incoming ray and the outgoing reflected ray is 80 degrees. What is the angle of incidence? [\[Light\]](#)
- (A) 50 degrees
 - (B) 100 degrees
 - (C) 20 degrees
 - (D) 160 degrees
 - (E) 45 degrees
 - (F) 40 degrees
- 18.** For the data 4, 7, 7, 2, 9, 7, 2, find the sum of the mode and the range. [\[Data Handling\]](#)
- (A) 2
 - (B) 11
 - (C) 9
 - (D) 14
 - (E) 16
 - (F) 7
 - (G) 21
- 19.** Why is letting untreated sewage flow into a river especially harmful to people downstream? [\[Wastewater Story\]](#)
- (A) it spreads disease-causing microbes and uses up the water's oxygen
 - (B) it makes the river flow faster
 - (C) it raises the river's salt to safe levels
 - (D) it has no real effect once diluted
 - (E) it adds useful minerals only
 - (F) it turns the water into drinking water

20. Two sides of a triangle are 7 cm and 10 cm. Which of these CANNOT be the length of the third side?

[The Triangle & its Properties]

- (A) 16 cm
- (B) 18 cm
- (C) 15 cm
- (D) 10 cm
- (E) 4 cm
- (F) 12 cm

21. A light ray makes an angle of 35 degrees with the flat surface of a plane mirror. Measured from the normal, what is the angle of reflection? [Light]

- (A) 35 degrees
- (B) 55 degrees
- (C) 65 degrees
- (D) 70 degrees
- (E) 45 degrees
- (F) 110 degrees
- (G) 25 degrees

22. The mean of six numbers is 14. A seventh number is added and the mean rises to 15. What is the seventh number? [Data Handling]

- (A) 21
- (B) 105
- (C) 16
- (D) 84
- (E) 20
- (F) 15
- (G) 18

23. A bar screen at a plant catches one large rag on average every 5 minutes. How many rags would it catch in 6 hours of steady flow? [Wastewater Story]

- (A) 144
- (B) 60
- (C) 720
- (D) 12
- (E) 72
- (F) 36
- (G) 30

24. An exterior angle of a triangle measures $4x$ degrees. Its two opposite interior angles are $(2x + 30)$ degrees and x degrees. Find the exterior angle. [The Triangle & its Properties]

- (A) 60 degrees
- (B) 40 degrees
- (C) 120 degrees
- (D) 100 degrees
- (E) 90 degrees
- (F) 150 degrees

25. Two plane mirrors form 7 images of a small bead. If the same mirrors are reset to a right angle, how many images will the bead now have? [\[Light\]](#)

- (A) 4
- (B) 11
- (C) 5
- (D) 7
- (E) 9
- (F) 3

26. Over two days, School A sold 120 then 150 tickets, while School B sold 90 then 200 tickets. By how many tickets did one school's two-day total beat the other's? [\[Data Handling\]](#)

- (A) 10
- (B) 30
- (C) 40
- (D) 20
- (E) 60
- (F) 50

27. Drinking water is disinfected with 0.5 mg of chlorine per litre. How many grams of chlorine are needed for 40,000 litres? Note 1 g equals 1000 mg. [\[Wastewater Story\]](#)

- (A) 200 g
- (B) 0.5 g
- (C) 20 g
- (D) 2000 g
- (E) 80 g
- (F) 2 g
- (G) 20000 g

28. In a right-angled triangle, the two non-right angles are in the ratio 1 : 2. What are these two angles? [\[The Triangle & its Properties\]](#)

- (A) 60 degrees and 90 degrees
- (B) 40 degrees and 50 degrees
- (C) 30 degrees and 90 degrees
- (D) 20 degrees and 40 degrees
- (E) 25 degrees and 50 degrees
- (F) 45 degrees and 45 degrees
- (G) 30 degrees and 60 degrees

29. When sunlight splits into its seven colours, they appear in a fixed order. Starting to count from the red end, which colour is the fifth one? [\[Light\]](#)

- (A) Yellow
- (B) Indigo
- (C) Red
- (D) Blue
- (E) Violet
- (F) Green

30. A fair six-faced die is rolled once. What is the probability of getting a number greater than 4? [\[Data Handling\]](#)

- (A) $\frac{5}{6}$
- (B) $\frac{1}{6}$
- (C) $\frac{1}{4}$
- (D) $\frac{1}{3}$
- (E) $\frac{1}{5}$
- (F) $\frac{1}{2}$
- (G) $\frac{2}{3}$

31. After full treatment, the cleaned water from a plant is often released to a river or used on fields. Why is it NOT usually sent straight to taps for drinking? [\[Wastewater Story\]](#)

- (A) it is too pure to drink safely
- (B) some microbes and dissolved matter may remain, so extra purification is needed first
- (C) rivers cannot carry clean water
- (D) fields cannot use clean water
- (E) drinking water must contain sewage smell
- (F) treated water is always poisonous forever
- (G) it would damage the pipes by being too clean

32. A median is drawn from a vertex of a triangle to the opposite side, which is 16 cm long. Into what two lengths does the foot of the median divide that side? [\[The Triangle & its Properties\]](#)

- (A) 9 cm and 7 cm
- (B) 16 cm and 0 cm
- (C) 12 cm and 4 cm
- (D) 8 cm and 16 cm
- (E) 10 cm and 6 cm
- (F) 8 cm and 8 cm
- (G) 4 cm and 12 cm

33. In the band of seven colours from sunlight, how many colours lie strictly between the red end and the violet end? [\[Light\]](#)

- (A) 1
- (B) 4
- (C) 7
- (D) 3
- (E) 5
- (F) 6

34. What is the mean of the first ten even numbers, that is 2, 4, 6, up to 20? [\[Data Handling\]](#)

- (A) 11
- (B) 22
- (C) 10
- (D) 110
- (E) 9
- (F) 12
- (G) 20

- 35.** Digesting sludge yields about 0.4 cubic metres of biogas for every kilogram. How much biogas comes from 500 kg of sludge? [\[Wastewater Story\]](#)
- (A) 40 m³
 - (B) 200 m³
 - (C) 500 m³
 - (D) 1250 m³
 - (E) 2000 m³
 - (F) 20 m³
 - (G) 125 m³
- 36.** A right triangle has hypotenuse 13 cm and one leg 5 cm. What is its area? [\[The Triangle & its Properties\]](#)
- (A) 12 cm²
 - (B) 32.5 cm²
 - (C) 30 cm²
 - (D) 65 cm²
 - (E) 84 cm²
 - (F) 60 cm²
 - (G) 30 cm
- 37.** A dentist holds a small concave mirror very close to a tooth, inside the mirror's focus. Compared with the real tooth, the image the dentist sees is... [\[Light\]](#)
- (A) larger, upright and virtual
 - (B) same size and real
 - (C) larger, inverted and real
 - (D) smaller, inverted and real
 - (E) same size, upright and virtual
 - (F) smaller, upright and virtual
- 38.** A data set has a range of 24. If its largest value goes up by 5 and its smallest value goes down by 3, what is the new range? [\[Data Handling\]](#)
- (A) 29
 - (B) 24
 - (C) 32
 - (D) 19
 - (E) 16
 - (F) 21
 - (G) 26
- 39.** A sewer delivers sewage at 3000 litres per minute into a holding tank that can take 90,000 litres. How long until the tank is full? [\[Wastewater Story\]](#)
- (A) 30 minutes
 - (B) 300 minutes
 - (C) 60 minutes
 - (D) 3 minutes
 - (E) 45 minutes
 - (F) 27 minutes
 - (G) 90 minutes

40. Two angles of a triangle measure 55 degrees and 65 degrees. What is the third angle, and what type of triangle is it? [\[The Triangle & its Properties\]](#)
- (A) 60 degrees, a right-angled triangle
 - (B) 50 degrees, a right-angled triangle
 - (C) 70 degrees, an acute-angled triangle
 - (D) 60 degrees, an obtuse-angled triangle
 - (E) 60 degrees, an acute-angled triangle
 - (F) 60 degrees, an equilateral triangle
 - (G) 120 degrees, an obtuse-angled triangle
41. A driver's side mirror is curved outwards, making it a convex mirror. Why is such a mirror chosen instead of a flat one? [\[Light\]](#)
- (A) it gives a smaller, upright image and a wider field of view
 - (B) it gives a larger image so objects look closer
 - (C) it makes an inverted real image of traffic
 - (D) it focuses sunlight to a point for safety
 - (E) it shows objects at their true size always
 - (F) it removes the need to look behind
 - (G) it turns the image upside down for clarity
42. How many modes does the data set 3, 3, 5, 5, 7 have? [\[Data Handling\]](#)
- (A) 0
 - (B) 2
 - (C) 7
 - (D) 1
 - (E) 3
 - (F) 5
43. In sewage, which of these is an example of an INORGANIC impurity rather than an organic one? [\[Wastewater Story\]](#)
- (A) dead leaves
 - (B) nitrates and phosphates from fertiliser
 - (C) food scraps
 - (D) paper bits
 - (E) oil and grease from cooking
 - (F) human waste
 - (G) vegetable peels
44. An exterior angle of a triangle is 130 degrees, and the two opposite interior angles are equal. What is the size of each of those interior angles? [\[The Triangle & its Properties\]](#)
- (A) 60 degrees
 - (B) 130 degrees
 - (C) 65 degrees
 - (D) 50 degrees
 - (E) 25 degrees
 - (F) 75 degrees

45. A simple periscope uses two plane mirrors, and each mirror turns the light through a right angle. Through what total angle has the light been turned between entering and leaving the periscope? [\[Light\]](#)

- (A) 180 degrees
- (B) 135 degrees
- (C) 90 degrees
- (D) 45 degrees
- (E) 0 degrees
- (F) 360 degrees

46. A class has 10 boys with mean marks 40 and 15 girls with mean marks 50. Find the mean marks of the whole class. [\[Data Handling\]](#)

- (A) 90
- (B) 48
- (C) 44
- (D) 47
- (E) 46
- (F) 43
- (G) 45

47. A twin-pit or composting toilet is recommended in places without sewers because... [\[Wastewater Story\]](#)

- (A) it must be emptied every single day
- (B) it uses more water than a flush toilet
- (C) waste turns into safe manure with no need for a treatment plant
- (D) it stores waste forever without change
- (E) it makes drinking water automatically
- (F) it sends waste directly into rivers

48. Two sides of a triangle are 5 cm and 9 cm. If the third side is a whole number of centimetres, how many different lengths are possible? [\[The Triangle & its Properties\]](#)

- (A) 5
- (B) 9
- (C) 8
- (D) 14
- (E) 13
- (F) 10
- (G) 4

49. An object stands 20 cm from a plane mirror. The mirror is pushed 6 cm towards the object. By how many centimetres does the object-to-image distance shrink? [\[Light\]](#)

- (A) 20 cm
- (B) 6 cm
- (C) 14 cm
- (D) 12 cm
- (E) 28 cm
- (F) 3 cm

50. The mean of eight numbers is 30. A ninth number equal to 30 is added. What is the new mean? [\[Data](#)

[Handling\]](#)

- (A) 30
- (B) 270
- (C) 33.3
- (D) 60
- (E) 31
- (F) 240
- (G) 27

51. A village has 250 houses, and 60% of them have a toilet. How many houses still have no toilet?

[\[Wastewater Story\]](#)

- (A) 90
- (B) 100
- (C) 40
- (D) 160
- (E) 25
- (F) 150

52. Each base angle of an isosceles triangle is 50 degrees. What is the vertex angle? [\[The Triangle & its Properties\]](#)

- (A) 130 degrees
- (B) 65 degrees
- (C) 40 degrees
- (D) 50 degrees
- (E) 30 degrees
- (F) 80 degrees
- (G) 100 degrees

53. Two plane mirrors meet at 36 degrees. A marble is placed between them. How many images of the marble are seen? [\[Light\]](#)

- (A) 5
- (B) 9
- (C) 36
- (D) 8
- (E) 11
- (F) 7
- (G) 10

54. Cards numbered 1 to 20 are shuffled and one is drawn. What is the probability that its number is a multiple of 3? [\[Data Handling\]](#)

- (A) $\frac{1}{5}$
- (B) $\frac{1}{4}$
- (C) $\frac{2}{5}$
- (D) $\frac{6}{10}$
- (E) $\frac{1}{3}$
- (F) $\frac{3}{10}$

55. Put these treatment steps in the order they actually happen: (i) chlorination, (ii) bar screens, (iii) aeration by bacteria, (iv) settling into sludge. [\[Wastewater Story\]](#)

- (A) aeration, then bar screens, then settling, then chlorination
- (B) aeration, then chlorination, then bar screens, then settling
- (C) chlorination, then aeration, then settling, then bar screens
- (D) settling, then bar screens, then chlorination, then aeration
- (E) bar screens, then settling, then aeration, then chlorination
- (F) bar screens, then chlorination, then settling, then aeration
- (G) chlorination, then bar screens, then aeration, then settling

56. The two perpendicular sides of a right triangle are 8 cm and 15 cm. How long is the hypotenuse? [\[The Triangle & its Properties\]](#)

- (A) 17 cm
- (B) 19 cm
- (C) 13 cm
- (D) 23 cm
- (E) 289 cm
- (F) 21 cm

57. Two plane mirrors at 30 degrees form images of small marbles. If 3 identical marbles are placed between them, how many images are seen in total (each marble giving the full count)? [\[Light\]](#)

- (A) 22
- (B) 30
- (C) 9
- (D) 36
- (E) 11
- (F) 33
- (G) 12

58. One set of 4 numbers has mean 12; another set of 6 numbers has mean 22. What is the mean of all 10 numbers? [\[Data Handling\]](#)

- (A) 18
- (B) 15
- (C) 20
- (D) 17
- (E) 19
- (F) 16

59. Wet sludge weighing 800 kg is 70% water by mass. What is the mass of the dry solids in it? [\[Wastewater Story\]](#)

- (A) 300 kg
- (B) 240 kg
- (C) 70 kg
- (D) 112 kg
- (E) 560 kg
- (F) 240 g

- 60.** The three angles of a triangle are x , $(x + 20)$ degrees and $(x + 40)$ degrees. What is the measure of the middle-sized angle? [\[The Triangle & its Properties\]](#)
- (A) 70 degrees
 - (B) 50 degrees
 - (C) 80 degrees
 - (D) 20 degrees
 - (E) 100 degrees
 - (F) 40 degrees
 - (G) 60 degrees
- 61.** A clear, upside-down image of a candle is caught on a paper screen. Which of these could have formed it? [\[Light\]](#)
- (A) a plane mirror
 - (B) a convex mirror placed close to the candle
 - (C) a plane mirror tilted at 45 degrees
 - (D) a convex (bulging) mirror
 - (E) any flat shiny surface
 - (F) a concave mirror with the candle beyond its focus
 - (G) a concave mirror with the candle inside its focus
- 62.** Three numbers are 12, 18 and an unknown value. If their mean is 16, find the unknown value. [\[Data Handling\]](#)
- (A) 16
 - (B) 12
 - (C) 20
 - (D) 18
 - (E) 14
 - (F) 22
- 63.** A tank holds 50,000 litres of sewage, of which 6% settles out as sludge. How many litres of clarified water are left above the sludge? [\[Wastewater Story\]](#)
- (A) 44000 litres
 - (B) 48800 litres
 - (C) 30000 litres
 - (D) 3000 litres
 - (E) 47000 litres
 - (F) 49400 litres
 - (G) 47 litres
- 64.** In which triangle does the median from a vertex always fall exactly along the altitude from that same vertex? [\[The Triangle & its Properties\]](#)
- (A) only in a triangle with a 90 degree base angle
 - (B) only in a scalene triangle
 - (C) when the two sides meeting at that vertex are equal
 - (D) only when one angle is obtuse
 - (E) only in a right-angled triangle
 - (F) in every triangle
 - (G) never

65. In a mirror a clock looks like it reads 9:15. What is the correct time? [\[Light\]](#)

- (A) 3:45
- (B) 2:45
- (C) 6:45
- (D) 8:45
- (E) 9:45
- (F) 2:35
- (G) 2:15

66. A bar graph shows rainfall of 80 mm in June and 100 mm in July. What is the percentage increase in rainfall from June to July? [\[Data Handling\]](#)

- (A) 2.5%
- (B) 125%
- (C) 12.5%
- (D) 80%
- (E) 20%
- (F) 25%
- (G) 100%

67. Each cubic metre of biogas can run a stove for 30 minutes. For how many HOURS can 200 cubic metres run the stove? [\[Wastewater Story\]](#)

- (A) 200 hours
- (B) 10 hours
- (C) 100 hours
- (D) 16 hours
- (E) 60 hours
- (F) 6000 hours
- (G) 50 hours

68. Two of the three exterior angles of a triangle, one at each vertex, are 120 degrees and 140 degrees. What is the third exterior angle? [\[The Triangle & its Properties\]](#)

- (A) 100 degrees
- (B) 40 degrees
- (C) 60 degrees
- (D) 80 degrees
- (E) 260 degrees
- (F) 160 degrees
- (G) 120 degrees

69. When the angle of incidence on a mirror is increased by 12 degrees, by how much does the angle between the incident and reflected rays change? [\[Light\]](#)

- (A) increases by 90 degrees
- (B) increases by 48 degrees
- (C) increases by 6 degrees
- (D) increases by 12 degrees
- (E) decreases by 12 degrees
- (F) increases by 24 degrees
- (G) stays the same

70. Nine test scores are 15, 22, 9, 30, 18, 27, 12, 25, 20. What is the median score? [\[Data Handling\]](#)

- (A) 25
- (B) 19.5
- (C) 20
- (D) 21
- (E) 22
- (F) 18
- (G) 9

71. At a huge fair, hundreds of mobile toilets are installed. The MAIN public-health reason is to... [\[Wastewater Story\]](#)

- (A) increase the fair's electricity supply
- (B) collect waste to sell as plastic
- (C) reduce the noise of crowds
- (D) provide drinking water to visitors
- (E) stop human waste from contaminating the ground and water and spreading disease
- (F) keep the air cool
- (G) make the fair look modern

72. Which of these side-length sets forms a right-angled triangle? [\[The Triangle & its Properties\]](#)

- (A) 4 cm, 5 cm, 6 cm
- (B) 5 cm, 7 cm, 9 cm
- (C) 6 cm, 8 cm, 10 cm
- (D) 5 cm, 6 cm, 8 cm
- (E) 6 cm, 7 cm, 9 cm
- (F) 2 cm, 3 cm, 4 cm

73. Held to a plane mirror, the word TOMATO is examined letter by letter. How many of its six letters appear unchanged after the left-right flip? [\[Light\]](#)

- (A) 1
- (B) 6
- (C) 2
- (D) 5
- (E) 4
- (F) 3
- (G) 0

74. In a quiz, three students scored 8 each, two scored 5 each, and one scored 14. What is the mean score of these six students? [\[Data Handling\]](#)

- (A) 48
- (B) 9
- (C) 10
- (D) 6
- (E) 8
- (F) 8.5
- (G) 7

- 75.** A plant receives 120,000 litres of sewage daily and must store 5% of it untreated each day. How many litres are stored over one week? [\[Wastewater Story\]](#)
- (A) 420000 litres
 - (B) 84000 litres
 - (C) 42000 litres
 - (D) 30000 litres
 - (E) 4200 litres
 - (F) 6000 litres
 - (G) 60000 litres
- 76.** An isosceles triangle has one angle of 100 degrees. What are the other two angles? [\[The Triangle & its Properties\]](#)
- (A) 45 degrees and 35 degrees
 - (B) 40 degrees and 40 degrees
 - (C) 40 degrees and 100 degrees
 - (D) 100 degrees and 100 degrees
 - (E) 50 degrees and 30 degrees
 - (F) 80 degrees and 0 degrees
- 77.** Two plane mirrors are placed exactly parallel and facing each other, with a coin between them. How many images of the coin form, in principle? [\[Light\]](#)
- (A) 3
 - (B) 2
 - (C) 11
 - (D) 360
 - (E) none
 - (F) 1
 - (G) infinitely many
- 78.** Tickets numbered 1 to 30 are placed in a box and one is drawn. What is the probability that the ticket shows a two-digit number? [\[Data Handling\]](#)
- (A) $\frac{2}{3}$
 - (B) $\frac{7}{10}$
 - (C) $\frac{9}{30}$
 - (D) $\frac{3}{10}$
 - (E) $\frac{21}{100}$
 - (F) $\frac{2}{5}$
 - (G) $\frac{1}{3}$
- 79.** A grit chamber slows the sewage flow so that sand, grit and small pebbles can settle out. What would happen to the plant's pumps and pipes if this step were skipped? [\[Wastewater Story\]](#)
- (A) the water would become cleaner faster
 - (B) the hard grit would scrape and clog the pumps and pipes
 - (C) the bacteria would multiply too fast
 - (D) the sludge would float instead of sink
 - (E) the chlorine would stop working
 - (F) nothing, grit dissolves in water

- 80.** Two sides of a triangle measure 7 cm and 10 cm, and the third side is a whole number of centimetres. What is the largest possible perimeter of this triangle? [\[The Triangle & its Properties\]](#)
- (A) 27 cm
 - (B) 17 cm
 - (C) 33 cm
 - (D) 23 cm
 - (E) 30 cm
 - (F) 24 cm
- 81.** In a torch, a tiny bulb is placed at the focus of a concave reflector. What does the reflector do to the light? [\[Light\]](#)
- (A) spreads it out in all directions
 - (B) shrinks it into a dot on the bulb
 - (C) sends it out as a straight parallel beam
 - (D) focuses it to a single far point
 - (E) turns it into a rainbow
 - (F) reflects only half of it
 - (G) makes it bend back to the bulb
- 82.** The mean of seven numbers is 20. The numbers 10 and 40 are both removed. What is the mean of the five that remain? [\[Data Handling\]](#)
- (A) 25
 - (B) 22.5
 - (C) 18
 - (D) 16
 - (E) 20
 - (F) 17
- 83.** From 50,000 litres of sewage, a plant produces 47,000 litres of clean water. Written in simplest form, what fraction of the sewage becomes clean water? [\[Wastewater Story\]](#)
- (A) $50/47$
 - (B) $6/50$
 - (C) $3/50$
 - (D) $23/25$
 - (E) $47/50$
 - (F) $94/100$
 - (G) $47/100$
- 84.** How many medians does a triangle have, and what is special about them? [\[The Triangle & its Properties\]](#)
- (A) four medians, one per corner and one centre
 - (B) three medians, and they all pass through one common point
 - (C) three medians that meet the sides at right angles always
 - (D) one median, at the longest side
 - (E) three medians, each parallel to a side
 - (F) three medians that never meet
 - (G) two medians, meeting at the apex

- 85.** A student measures the angle of incidence as 50 degrees and the angle of reflection as 40 degrees on a plane mirror. What is wrong? [\[Light\]](#)
- (A) nothing is wrong, they can differ
 - (B) the two angles must be equal, so a 40 degree reflection is impossible
 - (C) reflection must always be larger than incidence
 - (D) incidence must always be larger than reflection
 - (E) reflection is always half of incidence
 - (F) the angles must add to 90 degrees
 - (G) the angles must add to 180 degrees
- 86.** A shop sold shoe sizes as follows: size 6 sold 4 pairs, size 7 sold 9 pairs, size 8 sold 9 pairs, and size 9 sold 3 pairs. What is the mode of the shoe sizes sold? [\[Data Handling\]](#)
- (A) both 7 and 8
 - (B) 8 only
 - (C) 6 only
 - (D) 7 only
 - (E) there is no mode
 - (F) 9 only
 - (G) 25
- 87.** When lots of organic waste enters a pond, bacteria break it down and the fish soon die. What is the key reason the fish die? [\[Wastewater Story\]](#)
- (A) the water becomes too salty
 - (B) the fish forget how to swim
 - (C) sunlight is blocked so fish freeze
 - (D) the waste turns the water acidic instantly
 - (E) the bacteria use up the oxygen dissolved in the water
 - (F) the bacteria eat the fish directly
 - (G) the waste makes the water too cold
- 88.** A rectangle is 24 cm long and 7 cm wide. A diagonal cuts it into two right triangles. How long is the diagonal? [\[The Triangle & its Properties\]](#)
- (A) 23 cm
 - (B) 17 cm
 - (C) 625 cm
 - (D) 168 cm
 - (E) 26 cm
 - (F) 25 cm
- 89.** A jeweller sets two mirrors at 20 degrees to show many copies of a ring. How many images of the ring appear? [\[Light\]](#)
- (A) 17
 - (B) 19
 - (C) 20
 - (D) 9
 - (E) 16
 - (F) 18

90. A bakery made 1200, 1500, 900, 1800 and 1100 loaves on five days. By how many loaves did the best day exceed the daily mean? [\[Data Handling\]](#)

- (A) 1800
- (B) 400
- (C) 600
- (D) 1300
- (E) 500
- (F) 200

91. It costs 2 rupees to treat every 100 litres of sewage. What is the cost of treating 2,50,000 litres?

[\[Wastewater Story\]](#)

- (A) 5000 rupees
- (B) 250 rupees
- (C) 50 rupees
- (D) 50000 rupees
- (E) 2500 rupees
- (F) 1250 rupees
- (G) 500 rupees

92. A triangle has angles in the ratio 1 : 1 : 2. What is the measure of the largest angle, and what type of triangle is it? [\[The Triangle & its Properties\]](#)

- (A) 120 degrees, an obtuse triangle
- (B) 90 degrees, an obtuse triangle
- (C) 60 degrees, an equilateral triangle
- (D) 100 degrees, an obtuse triangle
- (E) 90 degrees, a right-angled triangle
- (F) 80 degrees, an acute triangle

93. You stand facing a plane mirror and step 2 cm closer to it. How much closer does your image come to you? [\[Light\]](#)

- (A) 1 cm
- (B) 3 cm
- (C) 4 cm
- (D) 2 cm
- (E) 0 cm
- (F) 8 cm

94. A bag holds 5 red, 3 blue and 2 green balls. One ball is drawn. What is the probability that it is NOT green? [\[Data Handling\]](#)

- (A) $\frac{8}{100}$
- (B) $\frac{3}{5}$
- (C) $\frac{1}{5}$
- (D) $\frac{7}{10}$
- (E) $\frac{4}{5}$
- (F) $\frac{1}{2}$
- (G) $\frac{2}{10}$

95. Of 80,000 litres of treated water, 30% is reused to water gardens and the rest goes to the river. How many litres reach the river? [\[Wastewater Story\]](#)

- (A) 8000 litres
- (B) 26000 litres
- (C) 24000 litres
- (D) 56000 litres
- (E) 50000 litres
- (F) 70000 litres
- (G) 56 litres

96. Why is an exterior angle of a triangle always greater than either of its two opposite interior angles? [\[The Triangle & its Properties\]](#)

- (A) because it equals one of them exactly
- (B) because all exterior angles are 90 degrees
- (C) because it equals 180 degrees minus their sum
- (D) because it equals their sum, so it must be larger than each one alone
- (E) because interior angles are always tiny
- (F) because it equals their difference
- (G) because exterior angles are always 120 degrees

97. Counting the seven sunlight colours from the red end gives one colour as the 4th; counting from the violet end also gives the 4th. Which single colour is it? [\[Light\]](#)

- (A) Green
- (B) Blue
- (C) Indigo
- (D) Orange
- (E) Yellow
- (F) Violet

98. The five numbers 4, 8, 15, 16, 23 have a median. If 42 is added to the list, what is the new median? [\[Data Handling\]](#)

- (A) 16.5
- (B) 15
- (C) 16
- (D) 20
- (E) 18
- (F) 23.5
- (G) 15.5

99. A plant takes in 100,000 litres of sewage. Screens and grit removal take out 2% as junk; then 8% of what remains settles as sludge. How many litres of sludge form? [\[Wastewater Story\]](#)

- (A) 2000 litres
- (B) 7840 litres
- (C) 7800 litres
- (D) 8000 litres
- (E) 10000 litres
- (F) 9800 litres
- (G) 7920 litres

100. A right triangle has perpendicular sides of 20 cm and 21 cm. What is its perimeter? [\[The Triangle & its Properties\]](#)

- (A) 41 cm
- (B) 29 cm
- (C) 70 cm
- (D) 841 cm
- (E) 50 cm
- (F) 61 cm
- (G) 62 cm